

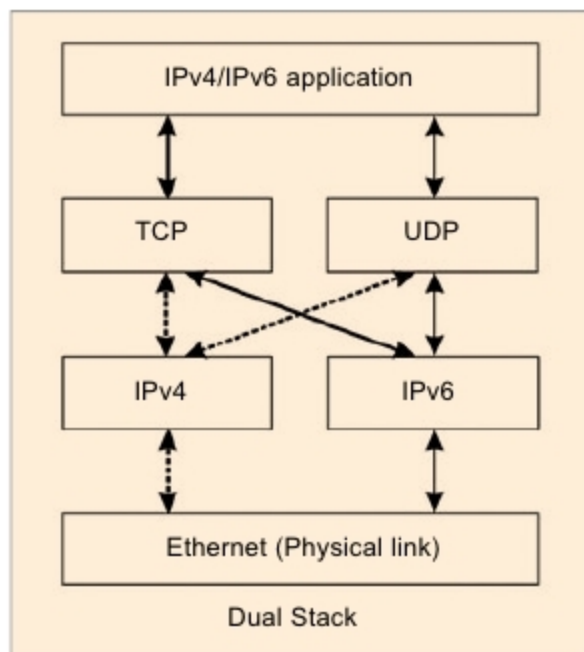


Fig. 1: Illustration of dual stack

In tunnelling, any node, when transmitting IPv6 datagram to IPv4 node, treats the whole IPv6 datagram as payload. With this payload, a new IPv4 datagram is made. IPv4 nodes treat all IPv6 packets as complete payloads for subsequent transmission in tunnelling. In the stated process, information of IPv6 datagram is preserved, but overhead increases. As a result, coding efficiency of the datagram, so framed, decreases. Thus, trade-off between dual stack and tunnelling is loss of information versus decreased coding efficiency.

However, as under tunnelling, IPv6 datagram is repackaged for transmission over IPv4, it is done with an encryption standard to hide the nature of the original datagram while running through the tunnel.

## Bootstrap protocol

Reverse address resolution protocol (RARP) is used to obtain the information of the IP address (logical address) corresponding to a given physical address of the machine. RARP does not provide the router or gateway address, server address, etc.

Bootstrap protocol (BOOTP) is used to provide additional information to a caller. BOOTP message goes as a payload of UDP layer. The message format is shown in Fig. 2. Fields in the message are as follows:

- Operation field. Set 1 and 2, respectively, for request and reply
- Hop field. Set to 0 in the request message. When the message is passed from one server to another, hop count is incremented by one

- Transaction ID field. Used for sequential co-ordination between request and reply messages, just like the sequence number in IP datagram
- Seconds field. Used to count time in seconds, since the host starts BOOTP

- Others field. Self-explanatory; vendor-specific fields are not yet standardised

When a device (say, computer) connected to a network is powered up and boots its operating system, the system generates a broadcast BOOTP message. The message is a request for an IP address. A BOOTP configuration server assigns the IP address based on request. With the start of BOOTP procedure, when a request is sent to the server, a timer is turned on. If no reply is received within the defined time period, BOOTP should attempt retransmission. BOOTP is implemented with UDP as transport protocol. It operates only with IPv4 networks.

## Dynamic host configuration protocol

IPv4 address is a stateful address. This means that, if a node moves from one subnet to another, the user has to either reconfigure the IP address or request for a new IP address from dynamic host configuration protocol (DHCP). DHCP is a network management protocol used over UDP/IP networks. A DHCP server dynamically assigns IP address and other network parameters to each of the devices connected to a network. With DHCP, an IP address is leased out to a device (host or computer) for a defined period of time.

|                                     |
|-------------------------------------|
| Operation (1 byte)                  |
| Hardware Type (1 byte)              |
| Hardware Length (1 byte)            |
| Hop (1 byte)                        |
| Transaction ID (4 bytes)            |
| Seconds (2 bytes)                   |
| Client IP Address (4 bytes)         |
| Your IP Address (4 bytes)           |
| Server IP Address (4 bytes)         |
| Router/Gateway IP Address (4 bytes) |
| Client Hardware Address (16 bytes)  |
| Server Host Name (64 bytes)         |
| BootFile Name (128 bytes)           |
| Vendor Specific Field (64 bytes)    |

Fig. 2: BOOTP message format

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### Ranges

| INPUT | RANGE (ANY ONE ONLY) |                   |
|-------|----------------------|-------------------|
| DC    | mV                   | 0 - 50            |
|       |                      | 0 - 60            |
|       |                      | 0 - 75            |
|       |                      | 0 - 100           |
|       |                      | 0 - 150           |
|       |                      | 0 - 200           |
| AC    | V                    | 0 - 20, 200, 1000 |
|       | mA                   | 0 - 2, 20, 200    |
|       | A                    | 0 - 2, 5          |
|       |                      | 0 - 20            |

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